

SECTION AAnswer **all** questions.

1. Using **outer** electrons only, draw a dot and cross diagram to show the formation of the bonding in sodium oxide. [2]

2. Complete the following sentence: [1]

Hydrogen bonding is an intermolecular force that occurs between molecules containing hydrogen atoms bonded to small, atoms which have lone pairs of electrons for example

3. By inserting arrows to represent electrons, show the electronic configuration of a calcium atom. [1]

1s	2s	2p	3s	3p	3d	4s

4. Hydrazine can be manufactured from ammonia.



Calculate the atom economy of this reaction. [1]

atom economy = %



5. The ammonium ion contains a 'coordinate bond'. Explain what is meant by this term. [1]

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6. Uranium is used in nuclear fuel reactors. One of its isotopes, uranium-235, has a half-life of 7.03×10^8 years and decays by α -emission.

- (a) Give the mass number and symbol of the element formed as a product of the radioactive decay of uranium-235. [1]

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- (b) If a quantity of uranium-235 decays, state what fraction is left after 2.812×10^9 years. [1]

fraction left =

7. Calculate the mass of calcium that contains the same number of atoms as there are molecules in 9.1 g of sulfur dioxide, SO_2 . [2]

calcium mass = g

